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A critique of John Hattie's theory of Visible Learning

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ABSTRACT

In this paper, I work out a five-stringed criticism of John Hattie's theory of Visible Learning. First, I argue that the theory is a theory of evaluation that denies education as such. Second, I show that there are problems with the dependent variable, learning, i.e. the effect of a given intervention. Thirdly, I show that Hattie's theory belongs to the radical constructivist paradigm. Thus, the problems of constructivism, i.e. problems of normativity and the outside world, walks directly into Hattie's concept of teaching, resulting in a double breakdown of the essence of teaching. Fourth, I argue that Hattie's concept of feedback has a centralizing trend which ultimately has the potential to transform the country's educational activities into a big hierarchical and data-driven organism. Finally, I show that Hattie's reference to Karl Popper's theory of "three worlds" is based on a highly problematic reading, where Popper's objective world is reduced to a subset of a radical subjectivity.

KEYWORDS

Learning; evaluation; Popper; Methodology; John Hatties

John Hattie's work has gained considerable influence in many countries, at both political and administrative levels, and across a variety of research communities. According to Scott Eacott, 'Hattie's work is everywhere in contemporary Australian school leadership' (Eacott, 2017). In the Scandinavian countries, we have a similar situation. Hattie's objective is to show what kind of interventions that stimulate learning outcome. He pursues this goal by pooling existing meta-evaluations into a synthesis of learning effects: a kind of double meta-evaluation. He constructs 138 independent variables, isolated from one another, and he studies their effect on learning outcomes, or on what is sometimes just called 'learning'. The effects of the 138 variables are measured by a learning effect, 'd', and the critical intersection point is a positive effect at $d > .40$. In its use of such a statistical approach to 'what works', Hattie's theory becomes an optimized version of the school effectiveness movement.

Hattie's empirical work has influenced many research groups, both educational and didactic. In some research communities Hattie's results are discussed in connection with the work of German empirical educationalists such as Andreas Helmke and Hilbert Meyer (von Oettingen, 2016; Laursen, 2016), and the empirical work has been both criticized and defended on methodological grounds (e.g. Hattie, 2010; Larsen, 2015; Snook, Clark, Harker, O'Neill, & O'Neill, 2010; Snook, O'Neill, Clark, O'Neill, & Openshaw, 2009).

However, the entire conceptual setup of Hattie's work—that is, the basic educational theory behind his empirical work—is discussed only rarely. That is certainly an omission, not least because Hattie himself insists that he is more interested in the educational narrative than in concrete empirical data

(Hattie, 2009, p. 22). Hattie's project is to build an entire educational idea and practice—an objective that has a corollary in his active involvement, with his students and consultants, in developing and selling educational concepts, for example the so-called 'Visible Learning +' and 'Challenging Learning'.

I therefore wish to analyze Hattie's theoretical basis. It is easy to articulate such a basis, because he devotes an entire chapter to the matter in his main work, *Visible Learning* (Hattie, 2009). The book's third chapter is even entitled 'The argument'. From this chapter, all concepts run through the entire book and into educational policy and practice.

Through separate chains of argument, I will make five interrelated and critical claims:

- (1) Hattie's theory is an example of what happens when a theory of evaluation marginalizes, forgets, and even destroys educational theory and practice.
- (2) There are major problems concerning the validity of the dependent variable: learning/achievement/learning outcome.
- (3) Hattie's theory is a double attack on the concept of teaching and on the European cultural tradition.
- (4) Hattie's concept of feedback has a built-in centralizing effect.
- (5) Hattie radicalizes an already reductionist reading of Karl Popper's 'three worlds' theory.

1. Hattie's theory is an example of what happens when a theory of evaluation marginalizes and even destroys or forgets educational theory and practice

Generally speaking, *Visible Learning* is not a learning theory in its own right, and in fact it is not a pedagogical theory at all. Rather, *Visible Learning* is what happens when education and learning are subjected to a rather simplistic theory of evaluation. Of course, Hattie's theory is also about measurement and behaviorism, but the point I want to make is that his theory requires all figures, data, and measures to be interpreted in line with concepts of from a rather simple evaluational scheme.

For the sake of the present argument, I define evaluation very traditionally as a retrospective investigation of the degree of realization of a given set of goals of a well-defined intervention, policy or program.¹ This means that goals, intervention, and assessment must be defined as separate and measurable entities that are not allowed to interact conceptually. In such a process we can focus on something that has already happened (summative evaluation) or on future improvements (formative evaluation). All other analytic terms must submit to this system.

I want to make four arguments for my claim that Hattie's theory is one of evaluation colonizing education. First, the overall goal of learning is that learners should be able to be 'self-monitoring, self-evaluative, self-assessing and self-learning' (Hattie, 2009, pp. 22, 37). The learner should even become a teacher: that is, an evaluative teacher who forces learners to set themselves through clear and measurable goals, goals that the learner will need to assess, evaluate and monitor in a maximizing and spiraling effect. Thus, the term 'evaluation' dominates even the most intimate pedagogical processes and relationships (Hattie, 2009, e.g. pp. 22, 24, 25 and 239). Evaluation is not something that happens after education; rather, education is simply defined by evaluation as such.

Second, this conclusion is reinforced by the opening chapters of Hattie's book, in which a number of key references refer not to educational theory but to theories of evaluation (e.g. Biggs, Walberg, Glass, Bloom, Cronbach, and Scriven). Furthermore, there are no systematic references to theories of education or didactics. Concepts such as subjects, world, opening, guidance, *Bildung*, democracy, authority, and discipline are nowhere to be found.²

Third, the evaluational approach is clear in what has become a kind of motto, 'Know thy impact', interpreted as a goal-driven and largely statistically process: that is, as evaluation theory. The motto states: 'Teachers become the learners of their own effect' (Hattie, 2009). This places education directly within the intellectual territory of evaluation theory. If that is true, we might expect, again, not that

education is something to be evaluated, but that education as such is sheer evaluation. The educational object has disappeared, one might say: it is transformed into sheer evaluation.

Thus, evaluation is at the core of Hattie's theory. This means that measurement, achievement, and intervention are the key categories that all educational concepts must adjust to. A short example is Hattie's understanding of teaching. Hattie says: 'This book focuses on student achievement ... Indeed, the role of teaching as intervention is developed throughout this book' (Hattie, 2009, p. 6). Notice how 'teaching' in an almost naive way is conceptualized as 'an intervention' addressing an isolated object of 'achievement', which was called 'effect' in the quotation just above.

Fourth, we should remember that Hattie's approach is not an ordinary evaluational theory, but rather a meta-evaluation approach. This means that Hattie does not connect to specific educational policies or educational goals. He merely summarizes a large number of evaluation reports that evaluate on the basis of many different situated objectives. This could be termed a sort of 'global assessment' in which evaluation is taken out of time and place and reconstructed as a global and digital structure. In Denmark, a group of researchers would call Hattie's meta-evaluation a 'pure evaluation', by analogy with the term 'pure education', which denotes an educational theory and practice that is claimed to work in self-regulating systems independent of time and space (Tanggaard, Rømer, and Brinkmann (2014)).

1.1. *The exclusion of education*

This point of departure results in a number of reductions of everything beyond the narrow horizon of evaluational theory. Here are a few examples, but in fact the rest of my article can be seen as an instance of this reduction.

Readers of Hattie's work might be seduced by a few short, sporadic, and friendly philosophical passages. He writes, for example, that the teacher is 'passionate and caring', and he quotes Aristotle as saying that the 'educated man, in every subject, looks for only so much precision as its nature permits' (Hattie, 2009, pp. 24 and 7). However, these occasional remarks are marginalized by Hattie's evaluation paradigm, which results in the following contradiction: one must be 'passionate, caring and not too precise', but at the same time learning objectives must be clearly formulated, measurable, and prepared for evaluation not only for the entire year, but for each lesson and each student. Hattie does not tell us how such an individualized, cognitive, and structuralist system of targets can be combined with social emotions or with Aristotle's sense of political and ethical community. This contradiction is the reason why there is no analysis or even further mention of any of these philosophical one-liners. Educational theory, therefore, ends up taking a repressive and ideological function, while the language game of evaluation rules on the level of the real.

Similarly, Hattie states that the teacher must have 'love of the discipline being taught' (Hattie, 2009, p. 24); but as was the case above, we hear no more about that in the rest of the book. The reason for this is that Hattie's teacher must see learning from the student's point of view and thus must install himself in the cognitive apparatus of the learner. The teacher must disappear into the learner, who can then SEE as a teacher (capitalization in original, p. 239). The teacher's 'love' for the profession and for the discipline becomes instead a passion and a love for learning, and thus in a way a passion for the teacher's self as it appears in the learner. What a discipline is, and what content is, is left hovering in the dark. Actually, in my view, Hattie's teacher must leave behind even the love of discipline, profession, science, and content, in favor of a self-love that is expressed in an idea of the student who tries to become an evaluational teacher. It is love for yourself in the other, without any sense of the other or indeed of your own self. In this conceptual process, both 'content' and 'the discipline' are completely marginalized.

In the book's final pages, educational theory does actually appear very briefly. Hattie imagines how Dutch theorist Gert Biesta would have responded to his ideas (Hattie, 2009 p. 254). Hattie says that Biesta thinks that educational 'efficiency' should be related to what is 'educationally desirable'. Hattie then states that 'achievement is among what is crucially desirable'. The 'discussion' is staged in just four lines. This, I think, is another example of the hostility to educational thought generated by the vocabulary of evaluation. Moreover, Biesta is called a 'she'.³

In the Danish research-based versions of Visible Learning, even these philosophical references, already very simple, disappear completely. In their introduction to Hattie's later book, *Visible Learning: For Teachers*, Niels Egelund and Lars Qvortrup understand educational theory analogously with medical science, and they strongly problematize that education should be a 'normative science', as they call it. Similarly, Hattie's own book starts with a long case study of modern cancer treatment (Hattie, 2009, p. viii). It is as if education corresponds to the treatment of serious illness. In my view, education is the opposite—a process of development and growth—but I will not go further into this. However, there is a close association between the medical sciences and the rise of evaluational theory, so perhaps the education–medicine analogy is not a coincidence.

Furthermore, Per Fibæk Laursen praises Hattie's work. Laursen thinks that educational thought has been built on what he calls 'misperceptions'—that is, reform pedagogy and conservatism, or, in short, theories of subjectivity and *Bildung* (see especially Laursen, 2009; but also Laursen, 2015, 2016). Laursen puts Hattie and evidence in the place of the rejected pedagogical traditions. So even though Laursen wants to restore an interest in 'the teacher', he is unable to do so, because as a consequence of Hattie's influence, he is caught within the discursive web of an evaluational vocabulary.

Let me give an example taken from Hattie himself of what happens when evaluation theory takes over. Hattie states that the independent variable 'school' has no particular effect on learning. But what is 'school' in the universe of evaluation theory? Well, it is, indeed, an evaluational *reconstruction* of a school. This means that a school consists of atomized elements, such as a school budget, the size of management, class sizes, etc., and that these 'elements' must be operationalized in such a way that they can all be placed in a statistical relation to a quantifiable 'learning outcome'. The concepts of both school and learning are first splintered, then reconfigured inside the rules of the language game of evaluation. Thus, learning becomes something outside 'school'. Learning becomes an effect, and school becomes an intervention/a method. We get the following evaluational benchmark method: combine the variables a , b , and c to get $d > .4$. In principle, all schools anywhere in the world could be the same.

Furthermore, Hattie's method results in a very specific view of what educational progress is. You might call it the ' $d > 0.4$ -method' (where ' d ' is a measure of the learning effect and ' 0.4 ' is the effect size that marks an improvement). Everything should be organized so as to circulate around the .4 figure. This particular figure becomes a kind of 'Lord of the Rings'. Here is an example of Hattie's idea of progress:

We can set benchmarks of what progress looks like (preferably $d = 0.40$ for every

- (3) Then Hattie returns to the tradition, armed with his statistical findings. But the tradition is now torn apart: it has become invisible to the eye of the evaluator. The tradition, along with the origins and the heart of the concepts and their corresponding educational practices, has become invisible. This invisible and divided residue the inner essence and movement of teachers must now perform as assessed on distinct and simple indicators that were originally a part of a historical and organic process.

Thus, evaluation theory has swallowed the very thing that it should evaluate. It then spits the remaining residues out as a poorly reflected logic of evaluation, a statistically and causally constructed system that is unable to see what it was supposed to evaluate.⁴

2. Problems of validity in the dependent variable

In my opinion, the most important methodological problem in Hattie's theory is the following. Even if we accept the idea of 'learning' as an evaluational effect, Hattie does not make a sufficiently precise definition of his dependent variable, that is, his effect variable: 'achievement outcome/learning outcomes/learning'. Furthermore, he tends to dismantle the significance of the independent variable, that is, of one of the 138 interventions, thereby causing a breakdown in the entire system of causality.

Actually, we do get a theoretical definition of learning outcome in chapter three. Hattie distinguishes between what he and many others in the post-Piagetian tradition call surface and deep learning, and he claims that these two levels often stand in opposition to one another. However, he also thinks that the contradiction can be reconciled in categories such as self-regulation, self-monitoring and self-evaluation, that is, in pure cognitive evaluational meta-skills that help the learner to become his own self-monitoring system. Hattie calls this system 'constructed understanding' (Hattie, 2009, pp. 27ff and 245).

Even though Hattie qualifies these distinctions with reference to Popper and Carl Bereiter, a point to which I shall return below, there is no real discussion of the conceptual levels and their relation to the empirical analysis. We are simply told too little about the difference between surface understanding, deep understanding and the constructed understanding. Bearing in mind that we are talking about the key effect variable, this is a serious criticism.

These theoretical uncertainties are reinforced in the empirical analysis, which is highly framed by the lack of conceptual clarity. No one knows, as far as I can see, how the dependent variable 'learning achievement' is operationalized, or whether it is surface learning, deep learning or constructed learning. Since we are talking about a synthesis of a meta-analysis of 60,000 evaluational findings, one must assume that there are many different operationalizations going on, but nobody knows, and therefore there is from the point of view of educational research no systematic relationship between theory and empirical data.⁵

A similar critique is made by one of Hattie's most influential advocates in Denmark, Lars Qvortrup (Qvortrup, 2015). Qvortrup thinks that Hattie's data focuses too much on 'surface learning'. Rather than drawing, as Hattie does, on Popper to qualify the concepts of surface and deep learning, Qvortrup tries to build a dependent effect variable based on Gregory Bateson's theory of learning. The reason why Qvortrup wants to avoid drawing on Popper is that Popper, in Qvortrup's view, is too philosophical. In the same spirit, Qvortrup believes that Hattie-inspired meta-analysis has replaced 'educational research based on arts and humanities' (my translation). Qvortrup even thinks that in the future, meta-analysis will itself be 'replaced by archives of raw data that allow the construction of complex data landscapes, illuminating the relationships between independent, dependent and mediating variables' (Qvortrup, 2015, p. 32, my translation). So we get the following sequence of decay: from the humanities, to meta-evaluation, to archives of raw data. This process points directly to the combination of technology and learning implicit in one of Hattie's passages quoted above.

In fact, it is the theory of evaluation that is active in these conceptual processes. Why? Because the evaluation approach demands an isolated and measurable operationalization of learning. But Popper's,

Bereiter's and indeed Bateson's theories are broad philosophical theories of culture and learning. This results in a tension between, on the one hand, demands for an evaluationally conceived dependent variable 'learning' and, on the other, a broad discussion of 'learning' drawn from science and philosophy. The theory of evaluation reduces the concept of learning, while science expands it. Since evaluation in this case rules, it is also the narrow evaluational conception of learning that determines the sequence of the language games.

In addition, another important critique of Hattie's dependent variable, i.e. 'learning outcomes/achievement', is that the evaluational concept of learning reduces scientific 'learning' in such a way that evaluational learning ultimately comes to define the independent variable, i.e. teaching. This makes it virtually impossible to establish any kind of real causality between intervention and outcome, the exact kind of relation that Hattie wants to explore. If that is correct, Hattie has constructed what I call a 'research cage': that is, a system with high reliability and internal consistency but devoid of any contact with the object as such because the terms are mutually and reciprocally defined by each other. The price of the high reliability is low validity and low external environmental correspondence. One loses track of the truth and the object of study. This, of course, is a variation on my point made in part 1B above.

3. Hattie's theory is a twofold attack on the concept of teaching and on the European cultural tradition

3.1. The dissolution of the teacher and the reconstruction of the teacher on the premise of dissolution

Although Hattie's book structurally is dominated by evaluational theory, he also briefly works with a number of distinctions taken from learning theory, which deserve to be assessed in their own right, that is, before they are exposed to the over-determining evaluational structure. My point is that the underlying theories of learning exclude both pedagogy and the external world. My further point is that Hattie's reconstruction of the teacher carries this exclusion with it. In summary, we get what I would call a twofold dissolution of the teacher.

Hattie's basic educational thesis, repeated again and again, sounds like this:

Learning / achievement / outcome / impact are maximized ...

'When teachers SEE learning through the eyes of the student', and 'when students SEE themselves as their own teachers'. (p. 238, capitalization in original; see also pp. 22, 24 and 25)

As I just stated, this will give us a new kind of dissolution of the teacher. Let me explain:

Hattie lays claim to a special kind of student-centerdness—the fact that the teacher must SEE learning through the student's eyes—and he states the following: 'It is not the knowledge or ideas, but the learner's construction of this knowledge and these ideas that is critical' (p. 239).

This is an example of classic constructivist anti-school thinking, in which interactions between teacher and culture ('knowledge and ideas') are reduced to student 'construction'. Hattie also disregards what he calls 'didactics' (Hattie, 2009, p. 25). Finally, he seems to connect these elements to the work of what Bereiter calls a 'container theory of consciousness'.

One would expect such a constructivist point of departure to lead directly to a passive teacher, but that is not the case. On the contrary, putting some of Hattie's words together, we get an 'excellent hyperactive learning maximizing $d > 0.4$ passion teacher'. This happens because the teacher, according to the visibility formula above, must integrate himself (i.e. his learning targets) into the student, so that the student can monitor himself from the teacher's point of view and so on. This drilling into the student's eye and mind, into his 'visibility', is in contrast to the classical constructivist teacher, who was a designer of learning so that the student could see and act for himself. Normally, the constructivist teacher is more or less absent from the learning process.

My thesis is that Hattie's teacher is constructivism's own reinvention of the teacher. If that is the case, then we get a double disintegration of the concept of the teacher. Hattie's teacher is not an anti-constructivist teacher, but an extension of constructivism itself into teaching: it is the constructivist teacher

as such. As a consequence, all the problems of constructivism—the problems with content, normativity, and the outside world—are imported directly into Hattie's reconceptualization of teacher and student.

Let me explain my more specific argument for this claim. The philosophical development of John Hattie's 'teacher' is rooted in a learning theory that is itself based on an attack on the concept of teaching. Therefore, Hattie constructs a teacher on the premise of a non-teacher. The outcome is a doubled non-teacher. Here is the argument:

In the 1990s, an important paradigm within learning theory was subjective constructivism, mainly represented by Ernst von Glasersfeld's *Radical Constructivism* (Glaserfeld, 1995). John Biggs' 'constructive alignment' and Ivar Bjørgen's term 'responsibility for own learning' were also part of this movement (Biggs & Tang, 2011 and Bjørgen, 2000). Both Glasersfeld and Bjørgen based their theory on individualistic and cognitive readings of different aspects of the psychological and philosophical tradition, and they had much in common. Bjørgen referred positively both to American pragmatism and to Rousseau's Emile in a kind of psychologically defined progressivism. Glasersfeld's interest in the matter was stimulated by the founder of solipsism, George Berkeley, and by readings of Kant that were completely without esthetic and normative aspects. Both theories were inspired by Piaget, and both writers accepted the solipsistic premise. Bjørgen could even exclaim: 'Learning must in the last instance be done by the individual himself—solipsistic!' (Bjørgen, 2000, p. 236, my translation). Therefore, all learning and knowledge are subjective constructions. The external environment, content, politics, and culture fade away and are often reconstructed in or reduced to an authoritarian teacher filling empty vessels, or to individually constructed learning targets. Knowledge is perceived as a static category, usually opposed to processes of learning. The consequence is a solipsistic turn in learning theory, in which the distance from a cultural and independent teacher and from schools, teaching and political theory could hardly be any greater.

Now, Hattie takes his point of departure exactly in Glasersfeld and related approaches. Hattie makes a peculiar argument. On the one hand, he recognizes Glasersfeld's concept of learning in full: 'The model of visible teaching and learning *combines rather than contrasts* teacher-centered teaching and student-centered learning and knowing' (Hattie, 2009, p. 26, my italics). So Hattie is not criticizing constructivism and solipsism as such. What Hattie, on the other hand, is unhappy with is that constructivism lacks a theory of teaching, i.e. a 'teacher-centered teaching'. Therefore Hattie can also say that concerning the teacher, his theory is 'almost directly opposite' to constructivism. In other words, Hattie wants a theory of teaching that acknowledges the constructivist learning in full, by expanding it. Bjørgen and Glasersfeld thought that they had a teaching theory: the teacher of learning designs. According to Hattie, they did not; they had only a learning theory. It is Hattie who will bring this mess to order. He wants to produce a radical constructivist theory in even fuller maturity than that of the 1990s. Hattie's theory therefore rests on a tradition that has its roots in a radical critique of the teacher as a cultural and professional authority. Hattie tries to construct, not a theory of a cultural and vocational teacher, but a theory of a teacher whose own dissolution is her basic and intrinsic premise.

That John Hattie's theory is a radical constructivist theory is evident in three ways. First, the outside world is completely taken out of Hattie's universe. Everything takes place in closed systems. There are no references to social, scientific, political or economic aspects of educational life. Second, constructivism appears in a mutual reinforcement between two philosophical movements. On the one hand, radical constructivism was launched as an attack on ontological and normative theories of teaching, and its philosophical outlook corresponded very well with the emerging Luhmann-inspired system theory whose proponents are usually very excited about Hattie's work.⁶ On the other hand, there is no room in Visible Learning for the current interest in the concept of *Bildung* and schooling and the accompanying discussions of educational critique and ontology. Third, constructivism can be identified in the fact that more traditional pedagogy, if mentioned at all, is put into the very simple and square categories of which I have already given some examples.

To illustrate the radical constructivism and Hattie's radicalization of learning theory, please take a look at the front cover of one of his books, which depicts sixteen separate brains. Inside each brain there are learning meters, with arrows pointing around $d = .4$. Just as with constructivism, we are inside the

head of the student; but remember, this is a corrected form of constructivism, because it is not really the learner, but the teacher who is inside the brain of the learner: the teacher who sees everything from the student's point of view, while the student sees everything from the teacher's point of view (Hattie & Yates, 2013).

Thus, Hattie's theory is an enforcement of radical constructivism. Hattie does not criticize constructivism itself. On the contrary, he accepts, recognizes, and co-opts it. He creates an educational theory that fits with the solipsism of constructivism.

There is virtually no social analysis in Hattie's book.⁷ This means that the self-monitoring brains can be taken into account by all sorts of interests. In Denmark, the brains are seized mainly by a mixture of system theory, 'competitive state theory', and some new versions of poststructuralism and even post-humanism. This may sound odd, but is actually easy to understand if one makes a detailed analysis of the philosophical connections (Rømer, 2015).

3.2. The break from Western tradition

Hattie's theory also has some wider implications, which arise in connection with the largely unconscious break from Western educational tradition. This becomes clear if we look at Hattie's understanding of activity and passivity.

Hattie has very little respect for any kind of 'passivity'. Here is a characteristic passage that praises some North American teachers: 'The activity was visible and 'in the air'; *passive was not a word in the vocabulary of these accomplished teachers*'. And we also hear that the teacher must be an 'activator' and a 'deliberate change agent' (Hattie in 2009, pp. 28–29, my italics).

This disregard

4. Hattie's concept of feedback has a built-in centralizing effect

There is one aspect of Hattie's work which, to my knowledge, is completely overlooked, even though it has significant consequences for educational practice. I am talking about his concept of feedback, which has a built-in centralizing effect.

According to Hattie, feedback is an extremely efficient method, offering an effect score of $d = .73$. But, like all his other concepts, Hattie's concept of feedback is philosophically unreflected. It has no connection with, for example, the dialog of Martin Buber or Mikhael Bakhtin, or with Dewey's 'transaction' or Socratic dialog or anything similar. Therefore I ask: What, then, is Hattie's concept of feedback, now that it is so philosophically empty? It consists of two aspects: a deductive and an inductive aspect, of which the inductive aspect is in fact even more deductive than its counterpart. The deductive aspect has something to do with how the student's behavior is directed into structures of operative goals and targets without the slightest pedagogical opening. Such a system can be found on a full-page figure in Hattie's book (Hattie, 2009, p. 176). Here we see how the technical operations of atomized teaching penetrate the system of learning, forcing the student to see himself as a self-assessing and self-monitoring learning unit; that is, in Hattie's conception, that 'the student sees himself as a teacher'.

But actually this is not the most important aspect of feedback. What is more significant, the inductive part, becomes clear in the following quote:

The mistake I was making was seeing feedback as something *teachers provide to students* (...). It was only when I discovered that feedback was most powerful when it is from *the student to the teacher* that I started to understand it better. (Hattie, 2009, p. 173, italics in original)

Thus, feedback is not primarily about making art, science, and culture visible to the student: it is more about making learning visible to the teacher. Feedback are bits of $d > .4$ information coming to the teacher from the learner, so that the teacher can design the learning situation more exact and with more surgical precision than before. Feedback is about 'making the teacher into a pupil of his own effects': about transforming the teacher into his own effect. It is the opposite of, for example, Richard Peters' 'education as initiation', an approach in which the student's appearance in the art, culture and values of society are considered to be the most important educational stuff.

This shift has a powerful centralizing effect. From this point in the argument, feedback is about collecting information and data from lower levels of learning. This takes place in the first instance from the tightly structured process of learning inside the eye of each student and in the class. Then this information is gathered by a reflective team of teachers, who afterwards produce strategies and methods that can maximize the d-score. This process is repeated up through the hierarchy. This means that the school principal must have information on d-scores from the teachers in the same way as the teacher should have information about the learners. The principal asks: which classes and what teachers score $d > .4$? The same process is then repeated at the municipal level: the municipal administration can control the schools on the same basic principle, that is, $d > .4$. And finally, the municipalities can be controlled by the state, or even the EU, in the exactly same way.

In this way, we end up with a nationally collected and centralized $d > .40$, which—digitally, with what Qvortrup calls 'raw data'—can control the entire educational life in society, including children's and families' private lives and habits. We end up with a global or a national system, including regional and local subsystems, that produces its own data from targets found in the eye of the learner. This data is used to confirm and consolidate the objectives in a huge computational big data system. We get a new post-human country in full visibility.

5. Hattie radicalizes an already reductionist reading of Popper's 'three world theory'

As already mentioned, Hattie builds his argument on an aspect of Popper's philosophy. In particular, Hattie takes Popper's 'three world theory' and combines it with his own concepts of surface, deep, and constructed learning.

In brief, Popper's three worlds are as follows (Popper, 2011):

World 1: The physical world

World 2: The subjective world

World 3: Cultural and scientific objects/theories

Now, Hattie relates World 1 to his concept of surface learning, World 2 to deep learning, and World 3, with a further reference to Carl Bereiter's work, to the knowledge society's demands for conceptual construction. According to Hattie, his educational project—the 'self-monitoring learner'—takes place on the level of World 3.

So Popper's World 3 is the explicit educational ideal for Hattie. But that is a problem, for the following reason. World 3 is the world of cultural and scientific objects that have nothing to do with learning, because learning, and solipsist learning in particular, is a property of World 2. There is simply no mention of World 3 objects in Hattie's books. Hattie forgets World 3, probably because of his evaluational and constructivist interests. Therefore, all objectivity simply collapses into subjective learning, that is, World 2.¹¹

Popper's World 3 is a world of objectivity. In his Tanner lecture, Popper states that World 3 consists of 'abstract objects such as great books or great theories or great symphonies', and he emphasizes that these objects are '*objectively great*' (Popper, 2011, p. 150, italics in original). But Hattie's world is a world of cognition: in other words, World 2. Here is no mention of any objective content. As already mentioned, the cover of one of Hattie's books, *Visible Learning and the Science of How we Learn*, is a drawing of sixteen brains, each with a meter inside. There is no objectivity or 'greatness' about that. It is a World 2. And surely, we are, with such a cover, not far from neurological processes of the brain, thereby ending up in World 1. The epistemological reason for this confusion is, I think, that Popper is a philosophical realist, while Hattie's theory is based on the opposite of realism, that is, a strong subjective constructivism.

Nevertheless, Hattie reconstructs Popper's World 3 as equivalent to a purely subjective world. It is strange. How does it happen?

Part of the answer is to be found in Hattie's frequent references to the book by the Canadian theorist Carl Bereiter, *Education and Mind in the Knowledge Age* (Bereiter, 2002). As his title indicates, Bereiter's book is a study of education and consciousness in a knowledge society. Like Hattie, Bereiter refers extensively to Popper, and it is this sequence of reference, Popper–Bereiter–Hattie, that is interesting. Already Bereiter makes a number of reductions of Popper's World 3. The main reduction in the present context is that Bereiter reduces what he calls 'cultural objects' to 'truths and untruths' (Bereiter, 2002, p. 75). He uses this reduction to withdraw large parts of the humanities and of theology from World 3 in a manner that is not consistent with the above quotes from the Tanner lectures. Popper often used Beethoven's symphonies as an example of a World 3 object. But with his strong emphasis on cognition and on 'knowledge society', Bereiter tends to exclude this possibility.

However, Hattie makes an even further reduction of Popper's ideas. After all, Bereiter still likes to talk about the independent work of scientific theories and objects, and he therefore maintains Popper's distinction between learning (which is a World 2 concept) and 'knowledge building', which he attributes to World 3 in what he even calls 'knowledge outside the mind': that is, the opposite of solipsism (Bereiter, 2002, p. 56). Thus, Bereiter's conception is still somewhat consistent with Popper's distinction between World 2 and World 3 although, as just stated, it is contrary to Popper's overall intention because Bereiter excludes humanities and culture, and therefore also many educational topics.

The distinction between World 2 and 3—between learning and knowledge—is finally lost in Hattie's work, which reduces objectivity in its entirety to learning. There are no references to teaching content or to cultural and scientific matters, only to 'achievements' that are always conceptualized as cognitive processes in evaluational systems: that is, World 2.

We end up in a highly unfortunate situation where Popper's theory loses its contact with cultural and scientific objects. Instead, a residual Popperian vocabulary—of 'criticism, bold conjectures, experimentation and falsification'—collapses into a matter of learning methods producing $d > .4$.

Popper's World 3 is converted into a World 2, and his realist theory of the things of culture and science and their critical foundation becomes a teaching theory for the exact opposite epistemological point

of view: for radical constructivism, solipsism, and cognition. We end up with a world without education and without Beethoven's symphonies, and also without things and scientific energy altogether.

6. Conclusion

In this paper, I have presented a five-stringed criticism of John Hattie's theory of Visible Learning. First, I have argued that the theory is primarily a theory of evaluation, which forgets and throws away educational concepts and relationships. Next, I have shown that there are problems with the dependent variable, learning: the effect of a given intervention. Learning is weakly conceptualized, and there is an unexamined tension between learning as it appears in evaluation theory and in educational theory. Third, I have shown that Hattie's theory belongs to the radical constructivist paradigm. Hattie attempts to add a constructivist concept of teaching to a constructivist concept of learning. Thus, the problems of constructivism—the problems of a lack of normativity and the outside world—walk straight into Hattie's concept of teaching. The result is a twofold breakdown of the essence of teaching. Fourth, I have argued that Hattie's concept of feedback inherently has a centralizing effect which ultimately has the potential to transform a country's educational activities into a large hierarchical and data-driven organism. Finally, I showed that Hattie's reference to Popper's theory of three worlds is based on a highly problematic reading in which Popper's objective world is reduced to its opposite.

Notes

1. Some research in evaluation is far more advanced than this simple definition indicates. However, Hattie does not attend to more philosophical or critical approaches at all (e.g. House & Howe, 1999).
2. See Terhart (2012), who criticizes Hattie for this oblivion concerning didactics and *Bildung*.
3. Biesta answers Hattie in Biesta (2015).
4. See Larsen (2015) for a similar critique. For a general discussion of the relationship between education and evidence see Rømer (2014).
5. Nielsen and Klitmøller (2017) investigate the research underlying Hattie's emphasis on 'feedback'. They find grave conceptual and methodological problems.
6. Luhmann also liked Glaserfeld's work (Luhmann, 1998).
7. Hattie actually admits this on the macro level (e.g. the effect of social class on learning outcome). However, he also completely dismisses other social questions.
8. See Rømer (2012) for elaboration on Deweyan passivity. A further example of recent interest in passivity is Vlieghe and Lewis (2017) who draws on Giorgio Agamben's concept of 'impotentiality'. See also Roth (2011) for a discussion of passivity based on Bakhtin and situated learning.
9. 'Practitioners often lack familiarity with evaluation and statistical methods ...' (Hattie, 2009, p. 253).
10. See Rømer (2003) for a discussion on the relationship between imagination and judgment.
11. Hattie's use of Popper and his relation to Carl Berietter is also discussed in Terhart (2012).

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